



Grade 8

FAST Mathematics

Sample Test Materials

The purpose of these sample test materials is to orient teachers and students to the types of paper-based FAST Mathematics questions. By using these materials, students will become familiar with the types of items and response formats they may see on a paper-based test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test. The sample test materials are not intended to guide classroom instruction.

Grade 8 FAST Mathematics Reference Sheet

Conversions within a System of Measure

Customary Conversions

1 foot = 12 inches
1 yard = 3 feet
1 mile = 5,280 feet
1 mile = 1,760 yards

1 cup = 8 fluid ounces
1 pint = 2 cups
1 quart = 2 pints
1 gallon = 4 quarts

1 pound = 16 ounces
1 ton = 2,000 pounds

Metric Conversions

1 meter = 100 centimeters
1 meter = 1000 millimeters
1 kilometer = 1000 meters

1 liter = 1000 milliliters
1 gram = 1000 milligrams
1 kilogram = 1000 grams

Time Conversions

1 minute = 60 seconds
1 hour = 60 minutes
1 day = 24 hours
1 week = 7 days
1 year = 365 days
1 year = 52 weeks

Conversions between Systems of Measure

Customary to Metric Conversion Approximations

1 inch = 2.54 centimeters
1 foot = 0.305 meters
1 mile = 1.61 kilometers

1 cup = 0.24 liters
1 gallon = 3.785 liters
1 ounce = 28.35 grams
1 pound = 0.454 kilograms

Metric to Customary Conversion Approximations

1 centimeter = 0.39 inches
1 meter = 3.28 feet
1 kilometer = 0.62 miles

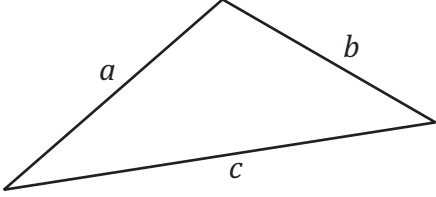
1 liter = 4.23 cups
1 liter = 0.264 gallons
1 gram = 0.0352 ounces
1 kilogram = 2.204 pounds

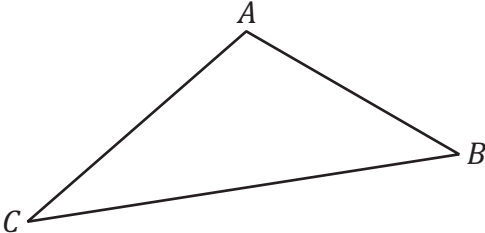
Formula

Slope Formula
$m = \frac{y_2 - y_1}{x_2 - x_1}$ where m = slope

Grade 8 FAST Mathematics Reference Sheet

Theorems

Triangle Inequality Theorem	
	$\begin{aligned}a + b &> c \\a + c &> b \\b + c &> a\end{aligned}$

Triangle Sum Theorem	
	$m\angle A + m\angle B + m\angle C = 180^\circ$

DO NOT REMOVE THIS SHEET

Use the space in this Test and Response Book to do your work. Then, completely fill in the bubble beside the answer you choose. For some items, filling in more than one bubble may be required, so read each item carefully. If you change your answer, be sure to erase completely.

Some items will ask you to write a response in a shaded box or boxes. See the sample item below.

Sample Item:

An expression is shown.

$$(22.4)(2.65)$$

What is the value of the expression?

Write your response in the shaded box below.

	Write your response in the box.
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Some items may have more than one box, so read each item carefully. Your answers for the items with response boxes may contain whole numbers, fractions, decimals, or negative numbers.

1. Four numbers are shown.

π	$\sqrt{8}$	$0.\overline{23}$	$\sqrt{49}$
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Complete the statement explaining which two numbers are rational.
For each box, fill in the bubble before the pair of numbers or word that is correct.

The two rational numbers are

- | |
|---|
| <p>☐ A π and $\sqrt{8}$</p> <p>☐ B π and $0.\overline{23}$</p> <p>☐ C $\sqrt{8}$ and $\sqrt{49}$</p> <p>☐ D $0.\overline{23}$ and $\sqrt{49}$</p> |
|---|

because they

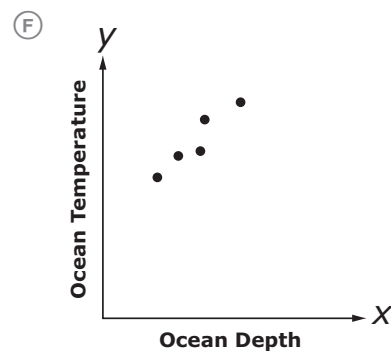
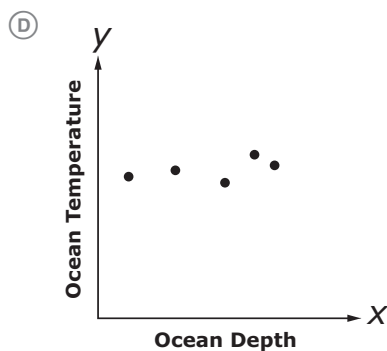
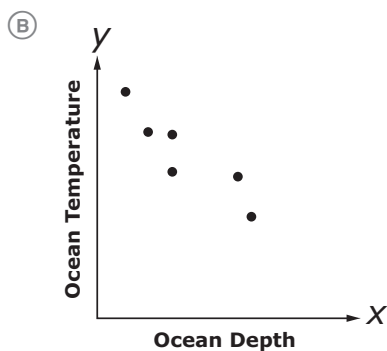
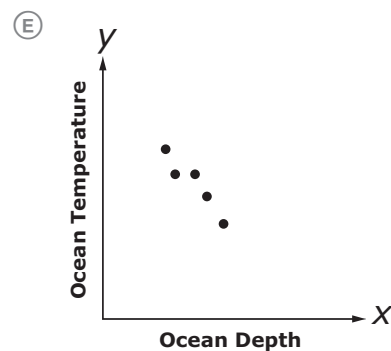
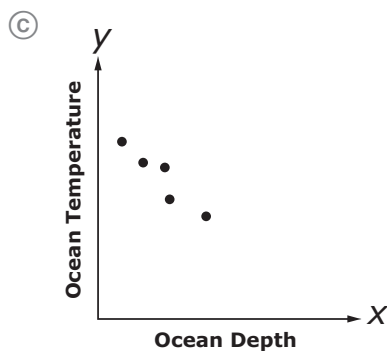
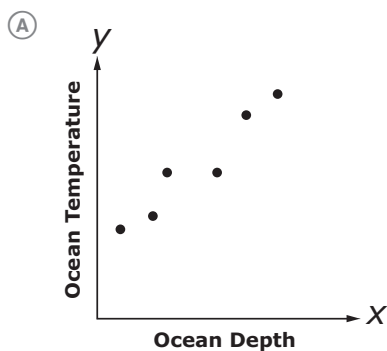
- | |
|--|
| <p>☐ A can</p> <p>☐ B cannot</p> |
|--|

be expressed as a ratio of two integers.

FAST Mathematics Sample Items

2. Gia records the temperature of ocean water at different depths and finds that the relationship between temperature and depth have a negative association.

Select all the scatter plots that could represent Gia's data.



3. An expression is shown.

$$\left(\frac{-1}{2}\right)^3 + \sqrt{8^2 - 15}$$

What is the value of the expression?

- Ⓐ $6\frac{7}{8}$
- Ⓑ $7\frac{1}{8}$
- Ⓒ $48\frac{7}{8}$
- Ⓓ $49\frac{1}{8}$

4. An expression is shown.

$$(2.3 \times 10^4) \times (4.1 \times 10^2)$$

Write your response in the shaded box below and fill in the bubble before the correct power of 10 to show the value of the expression in scientific notation.



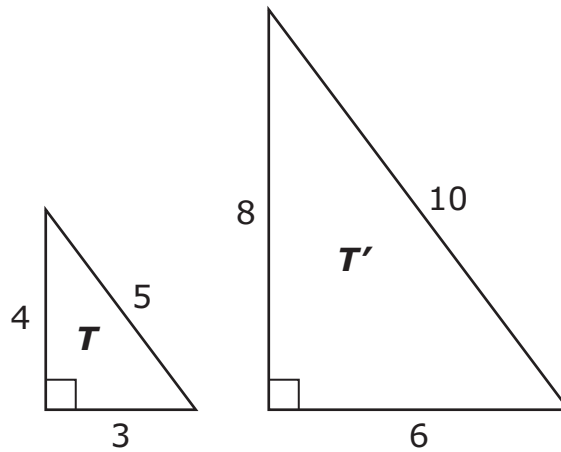
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- | | |
|-----|--------|
| (A) | 10^2 |
| (B) | 10^4 |
| (C) | 10^6 |
| (D) | 10^8 |

5. Fill in bubbles to identify the number of solutions to each equation.

	No Solutions	One Solution	Infinitely Many Solutions
$3x - 5 = 3(x - 2) + 1$	(A)	(B)	(C)
$3x - 5 = -9x + 15$	(D)	(E)	(F)
$3x - 5 = 3x + 10$	(G)	(H)	(I)

6. The transformation of triangle T to triangle T' is shown.



Select a word and a value to complete the description of the transformation. For each blank, fill in the bubble **before** the word or value that is correct.

Triangle T is _____ [☐ A congruent ☐ B similar] to triangle T' .

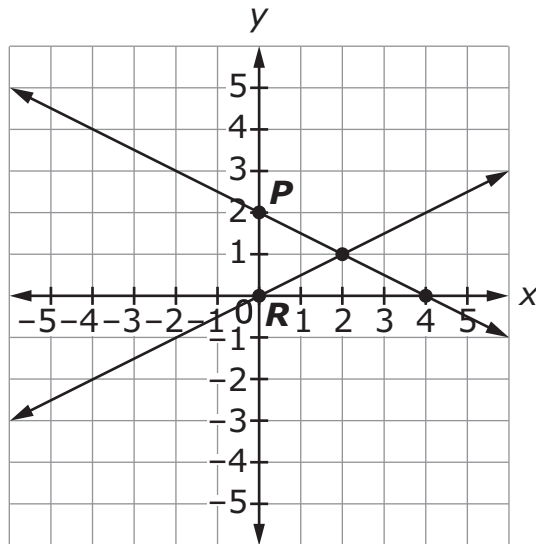
Triangle T can be transformed to triangle T' using a dilation with a scale factor of _____ [☐ A $\frac{1}{2}$ ☐ B 2 ☐ C 4].

7. A system of equations is shown.

$$y = \frac{1}{2}x$$

$$y = -\frac{1}{2}x + 2$$

Kayla graphs the system to solve it. She marks her solutions with P and R .



This question has **two** parts.

Part A

Which statement describes Kayla's solutions?

- Ⓐ Her solutions are correct.
- Ⓑ She marked the y -intercepts instead of the x -intercepts.
- Ⓒ She marked the y -intercepts instead of the intersection point of the two lines.
- Ⓓ She marked only the y -intercepts instead of the x - and y -intercepts.

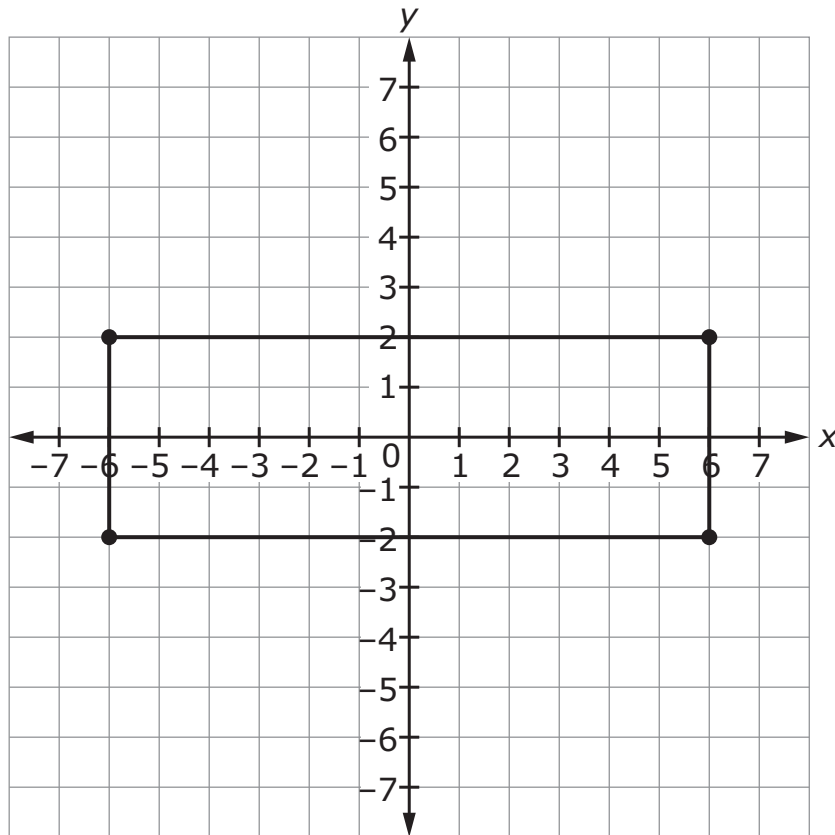
Part B

What is a solution to the system of equations?

Write your responses in the shaded boxes below.

( , )

8. A figure on a coordinate plane is shown.



Select all the transformations that will map the figure onto itself.

- Ⓐ translation up 4 units
- Ⓑ reflection across the y-axis
- Ⓒ dilation by a factor of 3 about the origin
- Ⓓ rotation 90 degrees clockwise about the origin
- Ⓔ rotation 180 degrees counterclockwise about the origin

- 9.** An expression is shown.

$$(1.3x)(2.6x + 4)$$

Write your responses in the shaded boxes below to complete the product of the expression.

	$x^2 +$		x
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Grade 8

FAST Mathematics

Sample Items

Answer Key

The Grade 8 FAST Mathematics Sample Items Answer Key provides the correct response(s) for each item on the sample test. The sample items and answers are not intended to demonstrate the length of the actual test, nor should student responses be used as an indicator of student performance on the actual test.

FAST Mathematics Sample Items Answer Key

1. Four numbers are shown.

π	$\sqrt{8}$	$0.\overline{23}$	$\sqrt{49}$
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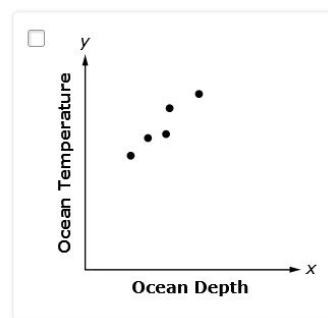
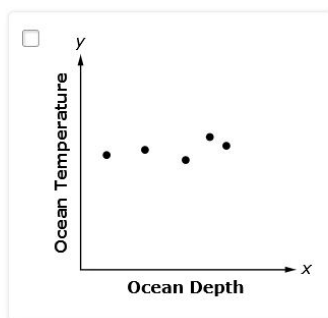
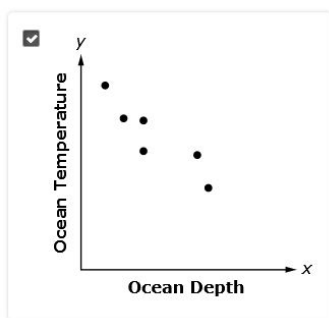
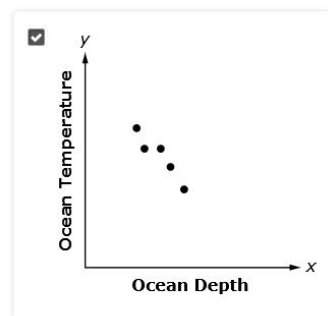
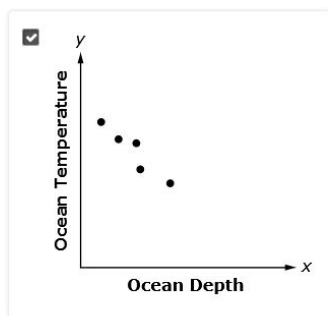
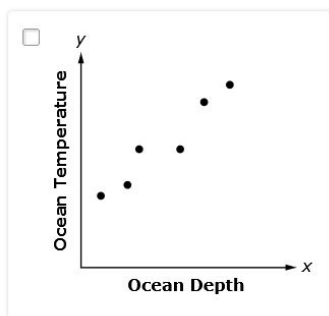
Complete the statement explaining which two numbers are rational.

The two rational numbers are $0.\overline{23}$ and $\sqrt{49}$ because they
can be expressed as a ratio of two integers.

FAST Mathematics Sample Items Answer Key

2. Gia records the temperature of ocean water at different depths and finds that the relationship between temperature and depth have a negative association.

Select all the scatter plots that could represent Gia's data.



Option B: **This answer is correct.** The student recognized that the scatter plot shows a negative relationship since the y -values generally tend to decrease as the x -values increase.

Option C: **This answer is correct.** The student recognized that the scatter plot shows a negative relationship since the y -values generally tend to decrease as the x -values increase.

Option E: **This answer is correct.** The student recognized that the scatter plot shows a negative relationship since the y -values generally tend to decrease as the x -values increase.

3. An expression is shown.

$$\left(-\frac{1}{2}\right)^3 + \sqrt{8^2 - 15}$$

What is the value of the expression?

☒ $6\frac{7}{8}$

☐ $7\frac{1}{8}$

☐ $48\frac{7}{8}$

☐ $49\frac{1}{8}$

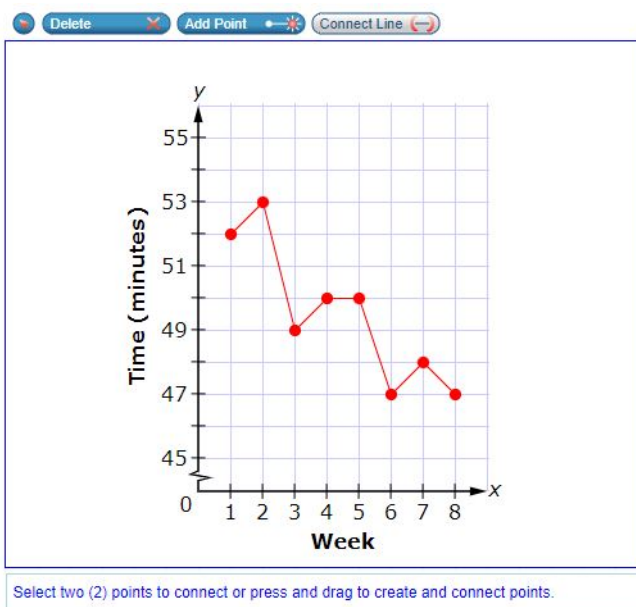
Option A: **This answer is correct.** The student correctly determined the value of the expression to be $\frac{55}{8}$ or $6\frac{7}{8}$.

FAST Mathematics Sample Items Answer Key

4. Louis tracks changes in his running time, in minutes, over a number of weeks. He records how long it takes him to run 5 kilometers once a week for 8 weeks. The table shows his data.

Week	Time (minutes)
1	52
2	53
3	49
4	50
5	50
6	47
7	48
8	47

Use the Connect Line tool to create a line graph of Louis's data.



FAST Mathematics Sample Items Answer Key

5. An expression is shown.

$$(2.3 \times 10^4) \times (4.1 \times 10^2)$$

Enter a number and select a power of 10 to show the value of the expression in scientific notation.

Other correct responses: any equivalent expression in scientific notation

FAST Mathematics Sample Items Answer Key

6. Select the boxes to identify the number of solutions to each equation.

	No Solutions	One Solution	Infinitely Many Solutions
$3x - 5 = 3(x - 2) + 1$	☐	☐	☒
$3x - 5 = -9x + 15$	☐	☒	☐
$3x - 5 = 3x + 10$	☒	☐	☐

FAST Mathematics Sample Items Answer Key

7. A table of values of a linear relationship is shown.

x	0	1	2
y	15	12	9

Create an equation to represent the relationship.

$y = -3x + 15$

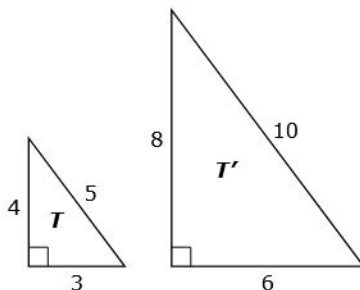
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1	2	3	x	y					
4	5	6	+	-	•	÷			
7	8	9	<	≤	=	≥	>		
	0		$\square^\square$	()		$\sqrt{\square}$	$\sqrt[\square]{\square}$	π	
.	-	$\frac{\square}{\square}$							

Other correct responses: any equivalent equation

FAST Mathematics Sample Items Answer Key

8. The transformation of triangle T to triangle T' is shown.



Select a word and a value to complete the description of the transformation.

Triangle T is to triangle T' .

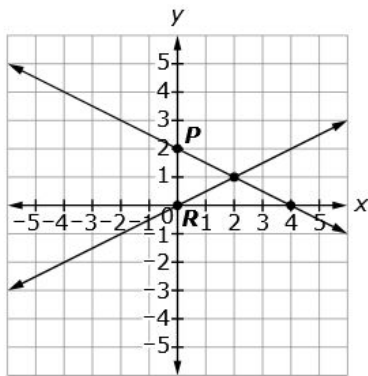
Triangle T can be transformed to triangle T' using a dilation with a scale factor of .

FAST Mathematics Sample Items Answer Key

9. A system of equations is shown.

$$y = \frac{1}{2}x$$
$$y = -\frac{1}{2}x + 2$$

Kayla graphs the system to solve it. She marks her solutions with P and R .



This question has **two** parts.

Part A

Which statement describes Kayla's solutions?

- ☐ A Her solutions are correct.
- ☐ B She marked the y -intercepts instead of the x -intercepts.
- ☒ C She marked the y -intercepts instead of the intersection point of the two lines.
- ☐ D She marked only the y -intercepts instead of the x - and y -intercepts.

Part B

What is a solution to the system of equations?

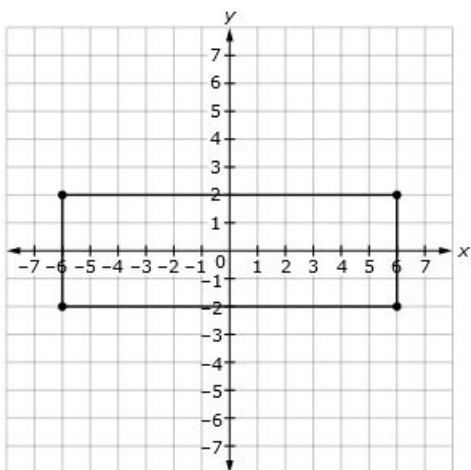
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Option C: **This answer is correct.** The student correctly identified that Kayla made a mistake and what her mistake was.

Other correct responses: for Part B, any equivalent values

FAST Mathematics Sample Items Answer Key

10. A figure on a coordinate plane is shown.



Select all the transformations that will map the figure onto itself.

- ☐ translation up 4 units
- ☒ reflection across the y -axis
- ☐ dilation by a factor of 3 about the origin
- ☐ rotation 90 degrees clockwise about the origin
- ☒ rotation 180 degrees counterclockwise about the origin

Option B: **This answer is correct.** The student identified that this reflection will carry the rectangle onto itself.

Option E: **This answer is correct.** The student identified that this rotation will carry the rectangle onto itself.

11. An expression is shown.

$$(1.3x)(2.6x + 4)$$

Enter values to complete the product of the expression.

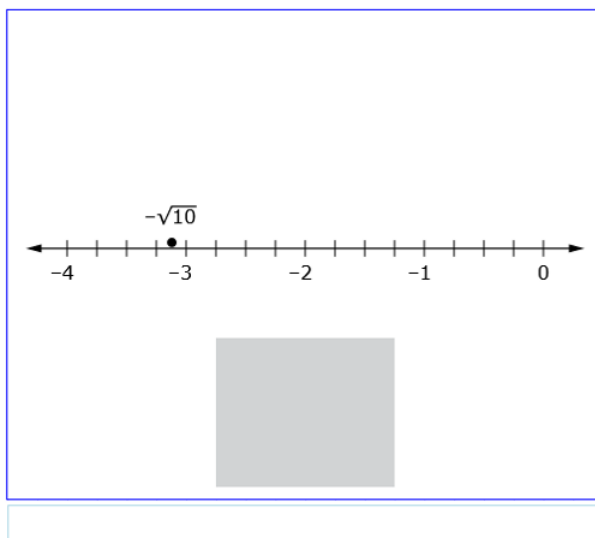
$$3.38 \text{ } x^2 + 5.2 \text{ } x$$

← → ↶ ↷ ✖		
1	2	3
4	5	6
7	8	9
	0	
.	-	$\frac{\Box}{\Box}$

Other correct responses: any equivalent values

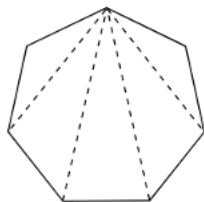
FAST Mathematics Sample Items Answer Key

- 12.** Drag the point onto the number line to show the approximate location of $-\sqrt{10}$.



Other correct responses: any value between -3 and -3.5 , exclusive

- 13.** A regular heptagon is separated into triangles, as shown.



Evan writes an incorrect expression to find the sum of the interior angles of the heptagon.

Evan's expression is shown. Click on **all** of the errors in Evan's expression.

$$(7 - 5) \times 90$$

FAST Mathematics Sample Items Answer Key

- 14.** An inequality is shown.

$$-\frac{5}{6}x - 12 \geq -17$$

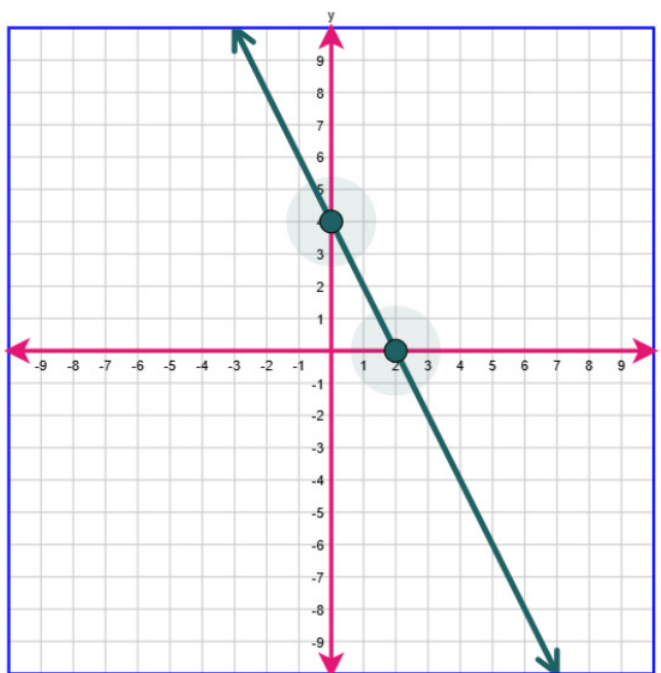
Create a graph on the number line to represent the solution to the inequality.



15. The values shown in the table represent a linear relationship.

x	y
-2	8
0	4
2	0
4	-4

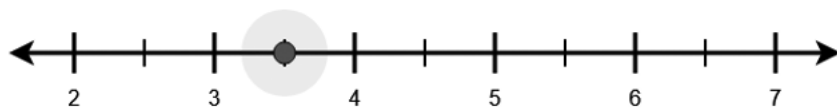
Create a graph of the relationship shown in the table.



Other correct responses: any line with a slope of -2 and a y-intercept of 4

FAST Mathematics Sample Items Answer Key

- 16.** Click on the number line to plot the approximate value of $\sqrt[3]{42}$.



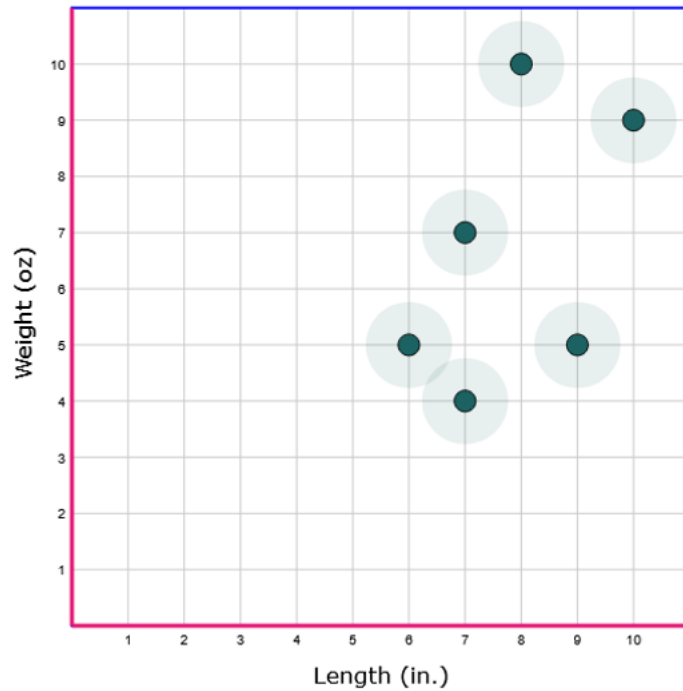
17.

Tony records the lengths, in inches (in.), and the weights, in ounces (oz), of six fish he catches. His data are shown in the table.

Fish Sizes

Length (in.)	Weight (oz)
8	10
7	7
7	4
10	9
9	5
6	5

Create a scatter plot of Tony's data.

Fish Sizes

18.

Janet records how many hours she practices the flute each week for 5 weeks. The table shows her data.

Flute Practice

Week	Time Practiced (hours)
1	2.0
2	3.5
3	2.5
4	4.0
5	3.5

Create a line graph of Janet's data.

Flute Practice

